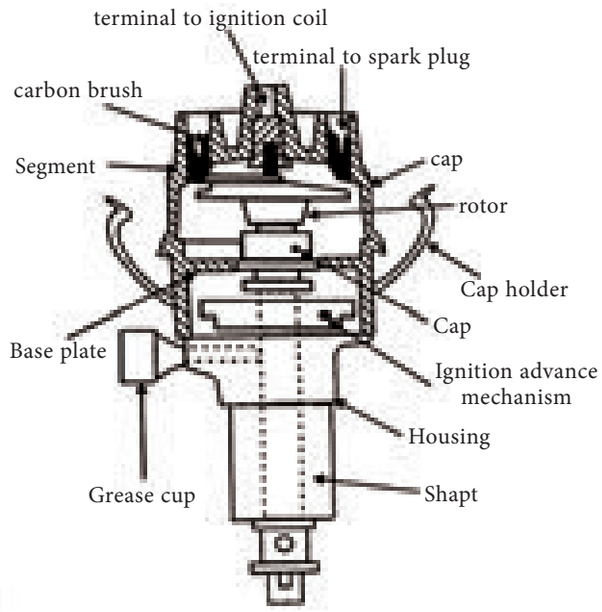


Figure 10.6 (a) Distributor



Figure 10.6 (b) Distributor

10.7 Ignition advance Mechanism

In an engine, the spark is timed just before the piston reaches TDC on the compression stroke. This will ensure the peak combustion pressure is obtained just after the TDC. If the pressure rise is before the TDC, then it will result in a weak power stroke. If the pressure rise is after the TDC, then the piston is moving down on the power stroke before combustion pressure reaches its maximum. When there is a change in the load and speed of the engine, then the ignition timing has to be adjusted accordingly to ensure maximum pressure just as the piston moves through the TDC and this is called as ignition advance.

10.7.1 Types of ignition advance

1. Vacuum advance mechanism
2. Manual method
3. Centrifugal advance mechanism
4. Combined centrifugal and vacuum advance

10.7.1.1 Vacuum advance mechanism

When the throttle valve is partly open, a partial vacuum develops in the intake manifold. The less air-fuel mixture gets inside the engine cylinder and hence fuel burns slower. The spark must be advanced at part throttle to give the mixture more time to burn.

The vacuum advance mechanism advances spark timing by shifting the position of the base plate. The vacuum advance unit has a diaphragm linked to the base plate. A vacuum passage connects the diaphragm to a port just above the closed throttle valve. When the throttle valve moves past the vacuum port, the intake manifold vacuum pulls on the diaphragm. This rotates the base plate so that the contact points open and close earlier.

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